NAME ____________________ DATE ____________________ PERIOD ____________________

ISS Greenhouse Module

Campaign 1 · Case 01 · Low Earth Orbit

SOLAR AGRICULTURAL AGENCY
Student Mission

MISSION QUESTION

Why are the lettuce roots tangled and directionless even though water, nutrients, temperature, and lighting appear normal?

SCIENCE FOCUS · GRAVITROPISM IN MICROGRAVITY

Plants normally use gravity to help guide root and shoot growth. In microgravity, that directional cue becomes unreliable, so engineers must provide another way for plants to orient.

Science status: The ISS case story is fictional. Gravitropism, statoliths, and plant responses to environmental cues are real.

REFERENCE

1 · Vocabulary

Gravitropism	Directional growth in response to gravity.	**Microgravity**	Apparent weightlessness during continuous free fall in orbit.
Statocyte	Plant cell involved in sensing gravity.	**Statolith**	Dense starch-filled structure that helps provide a direction cue.
Phototropism	Directional growth in response to light.		

PREDICTION

2 · Initial thinking

What evidence would help you distinguish a resource problem from an orientation problem?

Name the reading, observation, or record that would help.

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EVIDENCE TASK

3 · Investigate four evidence sources

Record only evidence that helps support or reject a cause. O = observation/data; I = inference/explanation.

Source	Observation or data	What does it support or reject?	O / I
Crew			
Sensors			
Plants			
Mission logs			

CAUSE TEST

4 · Test the competing explanations

Explanation	Status	One evidence-based reason
Nutrient solution is damaging roots.		
The light cycle is wrong.		
The seed stock is defective.		
Microgravity disrupts orientation.		

MECHANISM MODEL

5 · Build the mechanism

WORD BANK curve or grow without consistent orientation · downward · settle · settle in one direction

Earth gravity

Statoliths ______

→ consistent direction signal

→ roots grow ______

Microgravity

Statoliths do not ______ consistently

→ unreliable direction signal

→ roots ______

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DIAGNOSIS

6 · Diagnose and reject an alternative

State the cause in your own words.

Use the environmental condition and the plant mechanism.

Reject the most tempting alternative using evidence.

Name the alternative and the evidence that weakens it.

EXPLANATION

7 · Claim-Evidence-Reasoning

CLAIM	
EVIDENCE	
REASONING	

ENGINEERING RESPONSE

8 · Supply a consistent orientation cue

Choose or design root channels/mesh, a moisture gradient, shoot lighting plus a root guide, a rotating chamber, or another justified design.

Design	One criterion	One constraint

TRANSFER CHECK

9 · Exit ticket

Engineers add narrow root channels but do not change the lighting. Which symptom should improve first: tangled roots or pale leaves? Explain.

OPTIONAL EXTENSION

Compare a physical root guide with a rotating growth chamber. Which is easier to use on the ISS, and which more closely replaces gravity?

Student source line: NASA Plant Biology and Plant Gravity Perception resources; NGSS MS-ETS1 Engineering Design.